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IMPLEMENTATION COMPLETION AND RESULTS REPORT

IBRD LK-8145

ON A

LOAN

IN THE AMOUNT OF US\$213 MILLION

TO THE

Democratic Socialist Republic of Sri Lanka

FOR A

Metro Colombo Urban Development Project
December 14, 2022

Urban, Resilience And Land Global Practice
South Asia Region

CURRENCY EQUIVALENTS

(Exchange Rate Effective August 2022)

Currency Unit = LKR

LKR 365.65 = US\$1

FISCAL YEAR

January 1 – December 31

Regional Vice President: **Martin Raiser**

Country Director: **Faris H. Hadad-Zervos**

Regional Director: **John A. Roome**

Practice Manager: **Meskerem Brhane**

Task Team Leader(s): **Thalyta Ernandya Yuwono**

ICR Main Contributor: **Thalyta Ernandya Yuwono**

ABBREVIATIONS AND ACRONYMS

CBA	Cost Benefit Analysis
CMR	Colombo Metropolitan Region
CPF	Country Partnership Framework
CPS	Country Partnership Strategy
CMC	Colombo Municipal Council
EMF	Environmental Management Framework
ERD	External Resources Department
ERR	Economic Rate of Return
GDP	Gross Domestic Product
GRM	Grievance Redress Mechanism
GIS	Geospatial Information System
IFMS	Integrated Flood Management System
IRI	Intermediate Result Indicator
LA	Loan Agreement
LARC	Land Acquisition and Resettlement Committee
MCUDP	Metro Colombo Urban Development Project
MIS	Management Information System
MMWD	Ministry of Megapolis and Western Development
MoDUD	Ministry of Defense and Urban Development
MOF	Ministry of Finance
MUDH	Ministry of Urban Development and Housing
O&M	Operations and Maintenance
PAD	Project Appraisal Document
PCR	Physical Cultural Resources
PD	Project Director
PDO	Project Development Objective
PLA	Participating Local Authorities (Colombo Municipal Council, Sri Jayawedanapura-Kotte Municipal Council, Dehiwala-Mt. Lavinia Municipal Council, and Kolonnawa Urban Council)
PLI	PDO Level Indicator
PMMS	Project Monitoring and Management System
PMU	Project Management Unit
RAP	Resettlement Action Plan
RDA	Road Development Authority
RTC	Real Time Center
SCADA	Supervisory Control and Data Acquisition
SLLDC	Sri Lanka Land and Development Corporation
SLLRDC	Sri Lanka Land Reclamation and Development Corporation
SMF	Social Management Framework
UDA	Urban Development Authority

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**DATA SHEET****BASIC INFORMATION****Product Information**

Project ID	Project Name
P122735	Metro Colombo Urban Development Project
Country	Financing Instrument
Sri Lanka	Investment Project Financing
Original EA Category	Revised EA Category
Full Assessment (A)	Full Assessment (A)

Organizations

Borrower	Implementing Agency
Democratic Socialist Republic of Sri Lanka	Ministry of Urban Development and Housing

Project Development Objective (PDO)**Original PDO**

The project development objective (PDO) is to support the Borrower to (i) reduce flooding in the catchment of the Colombo WaterBasin, and (ii) strengthen the capacity of local authorities in the Colombo Metropolitan Area (CMA) to rehabilitate, improve and maintain local infrastructure and services through selected demonstration investments.



FINANCING

	Original Amount (US\$)	Revised Amount (US\$)	Actual Disbursed (US\$)
World Bank Financing			
IBRD-81450	213,000,000	199,059,117	199,059,117
Total	213,000,000	199,059,117	199,059,117
Non-World Bank Financing			
Borrower/Recipient	107,600,000	54,100,000	54,100,000
Total	107,600,000	54,100,000	54,100,000
Total Project Cost	320,600,000	253,159,117	253,159,117

KEY DATES

Approval	Effectiveness	MTR Review	Original Closing	Actual Closing
15-Mar-2012	10-Jul-2012	25-Nov-2014	31-Dec-2017	31-Dec-2021

RESTRUCTURING AND/OR ADDITIONAL FINANCING

Date(s)	Amount Disbursed (US\$M)	Key Revisions
23-Aug-2017	94.24	Change in Implementing Agency Change in Results Framework Change in Components and Cost Change in Loan Closing Date(s) Reallocation between Disbursement Categories Change in Institutional Arrangements Change in Implementation Schedule
24-Jun-2020	164.24	Change in Implementing Agency Change in Components and Cost Change in Loan Closing Date(s) Change in Implementation Schedule

KEY RATINGS

Outcome	Bank Performance	M&E Quality
Moderately Satisfactory	Moderately Satisfactory	Modest

**RATINGS OF PROJECT PERFORMANCE IN ISRs**

No.	Date ISR Archived	DO Rating	IP Rating	Actual Disbursements (US\$M)
01	27-Dec-2012	Satisfactory	Satisfactory	2.00
02	22-Jun-2013	Satisfactory	Satisfactory	34.12
03	16-Dec-2013	Satisfactory	Satisfactory	34.12
04	26-Jun-2014	Satisfactory	Moderately Satisfactory	52.27
05	04-Dec-2014	Satisfactory	Moderately Satisfactory	52.27
06	19-Jun-2015	Satisfactory	Moderately Satisfactory	65.89
07	12-Sep-2015	Satisfactory	Moderately Satisfactory	71.02
08	18-Apr-2016	Moderately Satisfactory	Moderately Satisfactory	82.78
09	11-Jul-2016	Moderately Unsatisfactory	Moderately Unsatisfactory	82.78
10	17-Feb-2017	Moderately Unsatisfactory	Moderately Unsatisfactory	94.24
11	23-Aug-2017	Moderately Unsatisfactory	Moderately Unsatisfactory	94.24
12	23-Mar-2018	Moderately Unsatisfactory	Moderately Unsatisfactory	94.24
13	27-Sep-2018	Moderately Unsatisfactory	Moderately Unsatisfactory	119.24
14	10-Apr-2019	Moderately Unsatisfactory	Moderately Unsatisfactory	129.24
15	23-Oct-2019	Moderately Unsatisfactory	Moderately Unsatisfactory	134.24
16	08-Dec-2019	Moderately Unsatisfactory	Moderately Unsatisfactory	134.24
17	02-May-2020	Moderately Unsatisfactory	Moderately Unsatisfactory	160.24
18	15-Dec-2020	Moderately Satisfactory	Moderately Satisfactory	176.24
19	30-Mar-2021	Moderately Satisfactory	Moderately Satisfactory	181.24
20	27-Sep-2021	Moderately Satisfactory	Moderately Satisfactory	189.24
21	26-Dec-2021	Moderately Satisfactory	Moderately Satisfactory	195.24



SECTORS AND THEMES

Sectors

Major Sector/Sector (%)

Public Administration 24

Central Government (Central Agencies) 7

Sub-National Government 6

Other Public Administration 11

Transportation 17

Urban Transport 6

Other Transportation 11

Water, Sanitation and Waste Management 59

Waste Management 6

Other Water Supply, Sanitation and Waste Management 53

Themes

Major Theme/ Theme (Level 2)/ Theme (Level 3) (%)

Finance 9

Finance for Development 9

Disaster Risk Finance 9

Urban and Rural Development 57

Urban Development 30

Urban Infrastructure and Service Delivery 30

Disaster Risk Management 27

Disaster Response and Recovery 9

Disaster Risk Reduction 9

Disaster Preparedness 9



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ADM STAFF

Role	At Approval	At ICR
Regional Vice President:	Isabel M. Guerrero	Martin Raiser
Country Director:	Diarietou Gaye	Faris H. Hadad-Zervos
Director:	John Henry Stein	John A. Roome
Practice Manager:	Ming Zhang	Meskerem Brhane
Task Team Leader(s):	Rosanna Nitti	Thalyta Ernandya Yuwono
ICR Contributing Author:		Thalyta Ernandya Yuwono



I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES

A. CONTEXT AT APPRAISAL

Context

1. Sri Lanka is a country of little more than 21 million people and in 2012 had a nominal gross domestic product (GDP) of US\$2,400 per capita. Economic density was (and is) concentrated in the Colombo Metropolitan Region (CMR), which accounted for almost half of the national GDP. The Government of Sri Lanka's (GoSL's) priority was to transform the Colombo Metropolitan Region (CMR) into an urban agglomeration that housed a modern world-class capital city through concerted efforts at economic, social, and physical regeneration. GoSL identified the key impediments to the realization of this envisaged transformation to be inadequacies in key infrastructure, including: (i) inadequate drainage capacity; (ii) *ad hoc* sewerage and solid waste disposal; and (iii) inefficient urban transport.
2. The Greater Colombo Water Basin comprises some 85 km² of land around Colombo. Floods in Colombo were caused by several major issues, including: (i) reduced storage capacity due to infilling of land and flood plain encroachments by illegal settlements; (ii) conveyance capacity of the canals being limited by solid waste, floating debris, and bottlenecks in the canals; and (iii) outflow capacity of the drainage system being too small, especially in the northern part of Colombo, where the New Mutwal tunnel outflow was reduced by rocks in the outlet. Severe flooding has taken place in Colombo in the last few decades, causing unprecedented socioeconomic losses: in June 1992, when rainfall of 494 mm was recorded in 19 hours; flooding in May 2010, when rainfall of 616 mm was recorded for nine days; and in November 2020, when 440 mm of rainfall was recorded in 16 hours¹. Repeated events of this nature are likely to be exacerbated by the effects of climate variability and were expected to cripple socio-economic activities of the most productive region of the country.
3. The project area covered the four Participating Local Authorities (PLAs) surrounding the Colombo Basin. The hydraulic infrastructure is formed by a micro-drainage system managed by Colombo Municipal Council (CMC), Sri Jayawerdanapura-Kotte Municipal Council, Kolonnawa Urban Council, and Dehiwala - Mount Lavinia Municipal Council (DMMC). The micro-drainage system consists of storm water drains that drain to the macro-drainage system of secondary and main canals, which prevents local inundations. Project interventions were expected to ensure that the Parliament area and its surroundings were kept within a safe water level at two-meter Median Sea Level (MSL), by enhancing drainage capacity by constructing new outfalls and upgrading existing outfalls, protecting canal banks and removing bottlenecks to maintain and improve the conveyance capacity of the canals. In addition, the project was to contribute to improving existing nature-based solutions in Colombo.
4. The project's high-level objective was aligned with two of the main strategic objectives of the World Bank's Sri Lanka Country Assistance Strategy (CAS) FY 09–12: improving infrastructure provision to strengthen competitiveness and enhancing the quality of services. Urban engagement was key to the CAS, which reflected the GoSL's post-conflict development strategy and priorities, as set out in GoSL's vision at appraisal. As the first Bank engagement in

¹ At appraisal, record shows that outlet capacity under extreme flood condition is estimated at 145 m³/s, equivalent to 147 mm/day of excess rainfall in the basin. Extreme rainfall in 1992 and 2010 were beyond the pumping and outlet capacity.

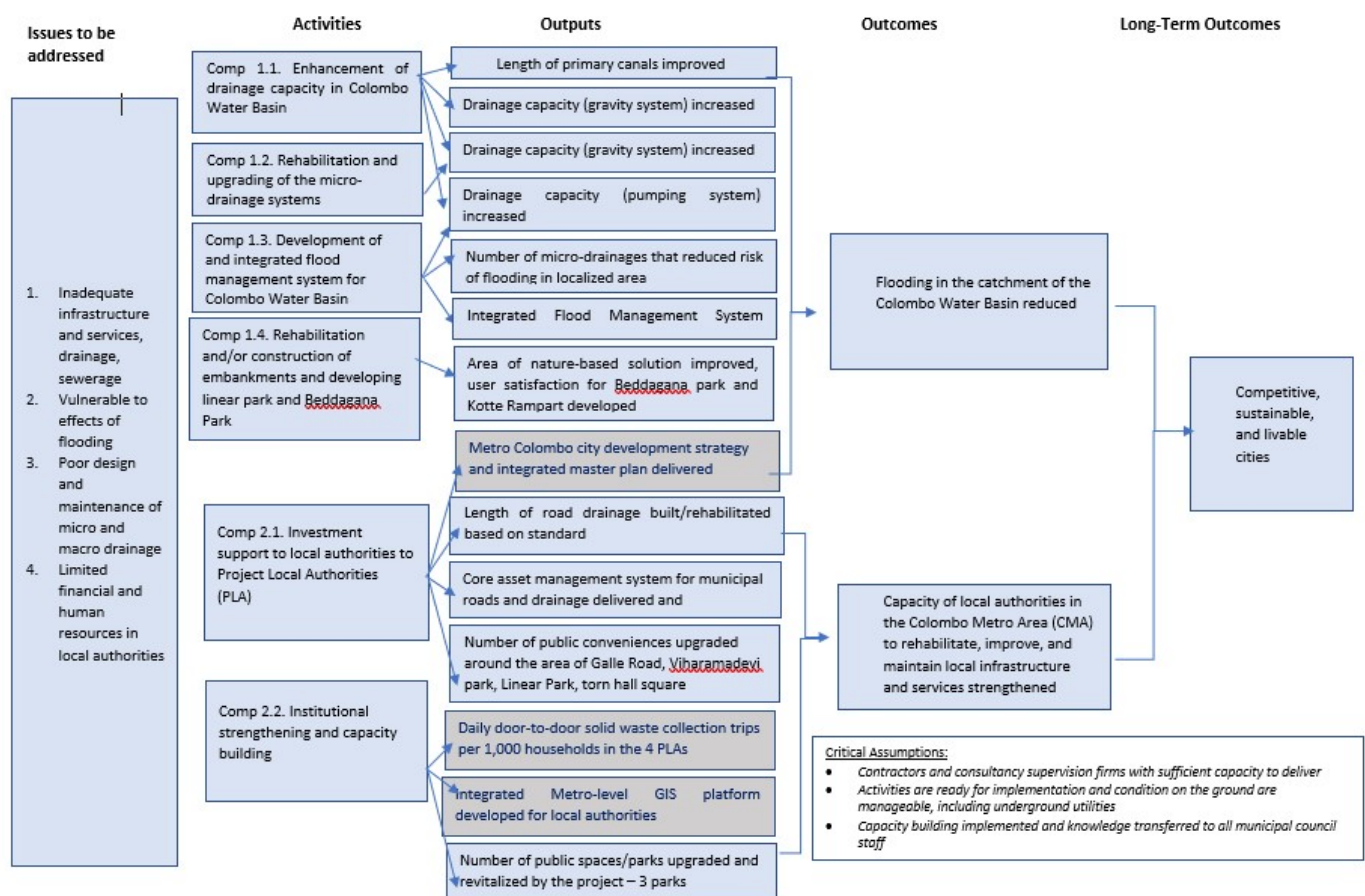


metropolitan Colombo, the project focused on critical priority infrastructure development, while also supporting the broader agenda of metropolitan development.

Theory of Change (Results Chain)

5. A Theory of Change (ToC) was not required at project appraisal. For the purposes of this ICR, the ToC in Figure 1 was derived from the project development objective (PDO), the project description, and the results framework in the Project Appraisal Document.

Figure 1. Theory of Change



Project Development Objectives (PDOs)

6. The PDO as stated in Schedule 1 of the Loan Agreement (8145-LK) is to: (i) reduce flooding in the catchment of the Colombo water basin; and ii) strengthen the capacity of local authorities in the Colombo Metro Area to rehabilitate, improve, and maintain local infrastructure and services through selected demonstration investments.



Key Expected Outcomes and Outcome Indicators

7. Two PDO level indicators were designed at appraisal to measure the achievement of project outcomes: (i) Reduction in the area under risk of flooding (50-year return period) in the project area; and (ii) Increase in percentage of total urban roads maintained by the four Participating Local Authorities (PLAs) that are in good and fair condition.

Components

8. **Component 1: Flood and drainage management in the Colombo Water Basin (US\$147.55 million).** This component comprised four sub-components: (i) *Sub-Component 1.1: Enhancement of drainage capacity in the Colombo Water Basin, including enhancement of runoff from the southeastern upper section of the basin, creation of lakes/retention areas in the central section of the basin, removal of bottlenecks in the downstream reaches of the canals, improvement of the outflow capacity and improvement/ construction of canal bank protections;* (ii) *Sub-Component 1.2: Rehabilitation and upgrading of the micro-drainage systems in priority flood-prone areas under the jurisdiction of Colombo Municipal Council;* (iii) *Sub-Component 1.3: Development of an integrated flood management system for the Colombo Water basin, including carrying out selected investments to improve public usability of canals and lakes;* and *Sub-Component 1.4: Rehabilitation and/or construction of embankments and developing a linear/nodal park along the east and southwest shores of Beira Lake and establishing a natural park around the degraded wetland in Beddagana.*
9. **Component 2: Urban development, infrastructure rehabilitation, and capacity building (US\$50.70 million).** *Sub-Component 2.1: Investment support to local authorities to enhance local capacity through implementing select high-priority infrastructure improvements in four local authorities in the Colombo Metropolitan Area to rehabilitate and improve drainage and roads, upgrade local public facilities, and purchase needed equipment to improve local public services. Sub-Component 2.2: Institutional strengthening and capacity building aimed to strengthen the capacity of the PLAs in road asset management, support metropolitan development strategies and planning.*
10. **Component 3: Implementation Support (US\$4.01 million).** Strengthening the capacity of the Borrower to carry out Project activities, including (i) provision of technical support to the Project management Unit (PMU) in the areas of Project management, monitoring and evaluation, procurement, financial management, and environmental and social safeguards; (ii) public awareness and communications support regarding Project interventions, public expectations, behavior changes and resettlement; (iii) provision of technical support to the Sri Lanka Land Reclamation Development Corporation (SLLRDC), Urban Development Authority (UDA), and the Participating Local Authorities (PLAs) in construction, supervision, and compliance with environmental and social safeguards; and (iii) acquisition of vehicles, office furniture and IT equipment for the PMU.

B. SIGNIFICANT CHANGES DURING IMPLEMENTATION

Revised PDO, PDO Indicators, and Outcome Targets

11. The PDO remained the same throughout the project cycle. However, the target of PDO indicator #1 was revised from two to three square km during the first restructuring in August 2017. This target was increased based on the updated technical analysis that 35 interventions under Component 1 were feasible for implementation and could be completed by the time of the extended closing date. PDO indicator #2 remained unchanged.



Revised Components

12. The Project Components remained unchanged.

Other Changes

13. As stated in the Loan Agreement, solid waste activities were part of Sub-Component 2.2. Institutional strengthening and capacity building for PLAs. The project was to provide capacity building and equipment for solid waste collection for the four PLAs. These were measured by intermediate indicators under Component 2. Following the collapse of Meethotamulla dump site in April 2017, GOSL requested the Bank to set aside approximately US\$7 million from Component 1 to finance a proper closure and stabilization of the site and other emergency measures at Meethotamulla, which were not part of the original project design. The Project provided advisory support and technical assistance for reviewing the feasibility study of a new sanitary landfill. Subsequently, the GoSL launched a separate program to address solid waste management in CMR and continued with the construction under GoSL funds outside MCUDP. As part of the first project restructuring, two new indicators on the number of public parks and public conveniences were introduced to replace the two indicators related to solid waste. MCUDP funds under Component 2 were utilized for the improvement of open spaces, including the townhall area and Viharamadevi Park.
14. Following the formation of the Ministry of Megapolis and Western Development (MMWD) in 2015, the GoSL requested the deletion of the proposed Metropolitan Colombo City Development Strategy (MCCDS) and the Integrated Master Plan from MCUDP. These activities along with the solid waste management (SWM) strategy were to be taken over by MMWD with funding from the GoSL budget. To ensure the strengthening capacity of local authorities continued, the Project incorporated the development of a Project Monitoring and Management System (PMMS) and provided equipment and training to PLAs and relevant agencies
15. At appraisal, the baselines and targets for some intermediate indicators were set based on the available information at appraisal, as the investments for macro and micro drainage, as well as their locations, had not been finalized. At the time of the Mid-Term Review, when the details of the investments and their locations were finalized, the baseline and the end target for drainage capacity (gravity system) were revised. Table 1 below summarizes the changes to intermediate indicators indicated above.

Table 1. Summary Changes to Intermediate Results Indicators

Original	New/Revised
Component 1	
IRI-1.2: Increase in drainage capacity (gravity system) – 268 m ³ /s	Based on technical analysis ² , end target was revised - 185 m ³ /s (part of the first restructuring in 2017)
IRI-1.4: Micro-drainage sub-projects implemented under the Project that reduce the risk of flooding in localized areas (10-year return period) - 15	Based on technical analysis and readiness for implementation within the project period ³ , the number of micro-drainage sub-projects implemented was revised to 6. However, the number of macro-drainages under the same component was increased.

² Hydraulic and hydrology modelling, as well as a LIDAR survey were conducted during initial years of implementation. The end target was adjusted and revised accordingly.

³ 15 micro-drainages were identified at appraisal in 2012. However, due to lack of funds during implementation and unforeseen underground utility issues, as part of the second restructuring in 2020, the end target was revised to six.



Component 2	
IRI-2.1: Metropolitan Colombo City Development Strategy, Integrated Master Plan delivered	Dropped. The Western Development Megapolis Plan was prepared by a separate agency. (Part of the first restructuring in 2017)
IRI-2.2: Km of road /drainage built / rehabilitated based on prescribed standards – 55	The target was changed to identified length of municipal roads feasible and eligible for rehabilitation (part of the first restructuring) – 45
IRI-2.3: Core asset management for municipal roads and drainage delivered, fully operational in the four PLAs – Y/N	Dropped. At the MTR it was agreed with the Road Development Authority that the project indicator should focus on the achievement of the result or output delivered to the four PLA. This is covered under IRI-2.2.
IRI-2.4: Households provided with access to regular solid waste collection in the 4 PLAs - % HH	Dropped, as discussed above (see paragraph 13)
IRI-2.5: Daily door-to-door solid waste collection trips per 1,000 households (HH) in the PLAs – number of HH	Dropped, as discussed above (see paragraph 13)
N/A	NEW: IRI-2.7: Number of public spaces/parks upgraded and revitalized by the project – three parks
N/A	NEW: IRI-2.8: # Public convenience maintained by PLAs that are in good and fair condition – 17 toilets

Rationale for Changes and Their Implication on the Original Theory of Change

16. The revisions to the results framework were primarily due to the following: (i) the decision by GoSL to first include activities related to the Meethotamulla dump site under MCUDP and then deciding to carry out these activities outside MCUDP; (ii) decision by GoSL to carry out some of the studies included in MCUDP by other Government agencies outside of MCUDP; (iii) updating the baselines and end targets of some of the intermediate indicators based on the results of the technical analysis that was carried out, as planned, during the initial years of project; (iv) addition of intermediate indicators to capture the outputs of invests in public parks and public conveniences, as well as the development of a Project Monitoring and Management System (PMMS), and to capture the outputs of capacity building for the PMU and the PLAs. These changes to the project indicators and their targets were consistent with the project objectives (of reducing flooding in the catchment of the Colombo water basin and strengthening the capacity of the local authorities in the Colombo Metro Area to rehabilitate, improve, and maintain local infrastructure and services) and did not affect either the scope or the scale of the project.

II. OUTCOME

A. RELEVANCE OF PDOs

Assessment of Relevance of PDOs and Rating

17. **The Project's objectives remained highly relevant at closing.** The Project is aligned with the World Bank Group's (WBG's) Sri Lanka Country Partnership Framework (CPF) for FY 17-FY20⁴. The PDO outcomes are fully aligned with the CPF's Engagement Area 3: Green Growth and Resilience and Objective 3.1. Improving urban livability, to develop urban areas in a more environmentally friendly manner.

⁴ At the time of the ICR, due to the impact of the COVID-19 pandemic and the macro and fiscal situation in Sri Lanka, CPF FY 17-20 was extended.



18. **The PDO remains highly relevant to the Government's priority in its development plan for Colombo.** The PDO and outcomes on improving local infrastructure in Metro Colombo and protecting CMR from flooding are aligned with the National Physical Planning Policy as well as Plan Sri Lanka 2011-2030. Specifically, the PDO contributes to the GoSL's objective of "ensuring that Colombo continues to perform its role as an influential regional economic centre." Urban wetland parks and improvement of public spaces complement the plans envisaged in the Colombo City Development Plan 2022-2030 prepared by the Urban Development Authority (UDA).
19. **The project remains relevant under the country's current economic crisis.** At closure, the country was in a difficult situation from the impact of COVID-19 pandemic and the start of the fiscal and political crisis. The PDO nevertheless remains highly relevant, as CMA is critical to the country's recovery and the PDO (reduce flooding in the catchment of the Colombo water basin and strengthen the capacity of local authorities in the Colombo Metro Area to rehabilitate, improve, and maintain local infrastructure and services) is crucial in this regard.

Relevance rating: High

B. ACHIEVEMENT OF PDOs (EFFICACY)

Assessment of Achievement of Each Objective/Outcome

Objective 1. To reduce flooding in the catchment of the Colombo Water Basin

20. **Summary.** The project achieved its objective to reduce flooding in the Colombo basin. Interventions implemented under the project were directed towards an integrated and system-wide physical infrastructure expansion supported by an automated decision support system for robust drainage management. The project outcome indicator related to reducing flooding and the intermediate result indicators related to Component 1 were fully achieved. With the completion of the flood mitigation investments, 322,400⁵ people in the Metro Colombo area are now protected from floods. In a recent survey, 81.49 percent of beneficiaries indicated that flood risks have declined.
21. **The project contributed to flood reduction even during high intensity rainfall.** The project successfully achieved its objective to reduce the area at risk of flooding around CMR for 3 km² surrounding the intervention areas from high intensity of rainfall of up to 50-year return period. Interventions under MCUDP contributed to allowable water levels in the canals of up to 2 meters above MSL, which ensures safe water levels around Colombo. The project contributed to improving the capacities of macro and micro drainage in Metro Colombo through: the construction of three pumping stations with a combined pumping capacity of 60 m³/s; the construction of two underground tunnels with a combined gravity drainage capacity of 43 m³/s; and the construction of two canal diversion projects with a combined gravity drainage capacity of 65 m³/s. Together with other improvements, the overall drainage system expansion resulted in the increase of total pumping capacity from 3.5 m³/s to 60 m³/s and the gravity drainage capacity from 100 m³/s to 220 m³/s. The macro drainage system is monitored and managed by an Integrated Flood Management System (IFMS), which was developed and installed as part of the project based on real time data to optimize stormwater drainage from the sub-catchments. The project covered the installation of individual⁶ and a central Supervisory Control and Data Acquisition system (SCADA), which is centrally controlled, and monitors flood gates and pumping stations constructed under the project. The IFMS shares information on hydro-meteorological data and

⁵ Number of beneficiaries protected from flooding was measured by measuring the total area of project interventions and its impact to populations living around the area up to 3km² from each intervention introduced under MCUDP.

⁶ Individual SCADA installed in all three pumping stations and flood gates, while the RTC housed the central SCADA.



forecasts with other national institutions for potential advance flood warning. In addition, information collected from this system enables the monitoring of the entire flood reduction system developed under the project, including the monitoring of the water level of the Kelani River.

22. **In the four Participating Local Authorities (PLAs)**, the project contributed to the improvement of local and micro drainage systems to alleviate flooding in isolated flooding hotspots in the city. The overall drainage interventions under the project have significantly reduced flooding in most parts of the Colombo Metro Region, benefitting both residents and daily migrants to the city. Within Colombo Municipal Council, six micro-drainage interventions to reduce local inundation were implemented, of which five have been completed; the Norris Canal intervention has been partially completed, with the final balance works being continued by the GOSL (see safeguards section of this ICR).
23. **Preservation of Colombo's natural wetland parks.** The project also contributed to technical and policy dialogue, which led to the preservation and improvement of natural wetland parks in the Metro Colombo area. This was integrated into the city's land use and led to Colombo receiving international recognition as the first capital to be recognized as Ramsar⁷ wetland city. Improvement of wetlands benefitted 50 percent of the metro-Colombo area through "natural air conditioning", as well as preserving 260 species and biodiversity in the Colombo wetlands.

Table 2. Indicators for Objective #1

PDO Indicator	End Target	Final Value	Level of achievement
Reduction in the area under risk of flooding (50-year return period) in the project area	3 km ²	3.03 km ²	Achieved
Intermediate Results Indicators			
Length of primary canals improved by the project	9.2 km	10.62 km	Exceeded target
Increase in drainage capacity (gravity system)	185 m ³ /s	220 m ³ /s	Exceeded target
Increase in drainage capacity (pumping system) subject to the results of the hydrological model	25 m ³ /s	60 m ³ /s	Exceeded target
Micro-drainage sub-projects implemented under the project that reduced risk of flooding in localized areas (10-years return period)	6	5	Almost Achieved
Real Time Center (RTC) developed and integrated into SLLDC operations	Yes	Yes	Achieved

Objective 2. To strengthen the capacity of local authorities in the Colombo Metropolitan Region to rehabilitate, improve and maintain local infrastructure and services through selected demonstration investments.

24. **Summary.** The project achieved its objective to strengthen local capacity within Colombo metro area on improving and maintaining local infrastructure and services. Results from a beneficiary survey indicated that almost 80 percent of the people in Metro Colombo are satisfied with the improvements to traffic flow, pedestrian walks, and public open spaces/parks for the elderly and children.

⁷ The **Ramsar Convention**, signed in 1971 in Ramsar, Iran, is the only global treaty that focuses specifically on wetlands. Today 170 nations are signatories to the Ramsar Convention. A contracting party agrees to nominate at least one wetland in its territory to the List of Wetlands of International Importance based on enumerated criteria.



25. **To achieve this objective, the project trained staff in the four PLAs and the MCUDP implementing agencies on operating machinery and equipment to operate and maintain local infrastructure.** Training focused on purchasing equipment to improve Operations and Maintenance (O&M) capacity, such as: (i) equipment to maintain the Talangama tank in Sri Jayawerdana-Kotte Municipal Council and improve the canal surrounding the area; (ii) machinery for the improvement and maintenance of roads in Colombo Municipal Council; (iii) dredging equipment for canal O&M; and (iv) waste collection equipment. In addition, the project provided hardware and software to improve M&E and O&M in the four PLAs, i.e., the purchase of MIKE urban software, MIKE II software for hydraulic modelling, and ArcGIS. With support from a parallel Bank Executed Trust Fund (BETF) activity on the Metro Colombo Transformation Platform (MCTP), MCUDP PLAs were provided training on urban Geospatial Information System (GIS) and Monitoring and Evaluation (M&E). Following the completion of MCTP and MCUDP, the local authorities are now providing funds under their budgets to maintain the software and data integration.
26. **The project successfully improved the physical aspects and the use of 50.8 hectares of the Colombo Townsquare area,** including the Central Business District and the National Hospital, where around 60,000 people pass by daily and use this urban public space. The interventions at Town Hall Square aimed to retain the existing architectural and environmental character of the area, while improving access to all public areas located within the Town Hall Square, including improvement of the Viharamadevi Park Interventions included better pedestrian footpaths and a bicycle lane through the park to improve connectivity and enhance recreational and fitness use of the park, as well as 2.36 km of walkability around the townhall. In addition, improved existing public toilets in 16 selected locations contributed to more hygienic and convenient sanitation facilities for the public, including daily commuters, domestic and foreign tourists. As part of integrated urban-upgrading programs, these facilities are in Galle Road, Beira Lake Linear Park, Town Hall Square, Viharamadevi Park, and within the network of roads and pedestrian walkways that were upgraded under the project.
27. **Within the four PLAs⁸, the project successfully rehabilitated and/or built 47 km of roads and pedestrian walks, and more than 70 percent of these roads are in good or excellent condition⁹.** The project contributed to the rehabilitation of two key arteries: Galle Road, the main arterial road in Colombo and its commercial spine, and a parallel R.A. De Mel Mawatha (Duplication) Road with a large commercial development. Both roads contribute to the flow of vehicular traffic into and exiting the city, especially during peak hours. The improvement of storm water drainage along 1.2 km of Vauxhall Street reduced flooding and contributed to the overall drainage improvement of the Colombo Metro area, while replacement of the gulley and minor repairs improved the flow of storm water along existing drainage lines. The project also asphalted 19 streets parallel to sidewalks.
28. A beneficiary survey conducted for the three public parks, including the wetland Beddagana Park, indicated that 78.6 percent of beneficiaries are satisfied with the urban space improvements and highlighted the positive change in the environment in Colombo over the past ten years in terms of pedestrian safety following the improvement of the roads and sidewalks. A total of 76.7 percent indicated that their land values increased because of the urban improvements over the last few years.

⁸ Four PLAs: Colombo Municipal Council (CMC), Sri Jayawerdanapura-Kotte Municipal Council, Dehiwala Mt. Lavinia Municipal Council, and Kolonnawa Urban Council.

⁹ 'Road in fair and good condition' in Sri Lanka is measured by International Roughness Index (IRI) between 3 and 7.



Table 3. Indicators for Objective #2

PDO Indicator	End Target	Final Value	Level of achievement
Increase in percentage of total urban roads maintained by four PLAs that are in good and fair condition	70	70	Achieved
Intermediate Results Indicators			
Km of road/drainage built/rehabilitated based on prescribed standards	45	47	Exceeded Target
Users satisfied with area-based demonstration projects in Colombo city	80%	78.6%	Achieved
Number of public spaces/parks upgraded and revitalized by the project	3	3	Achieved
Public convenience complexes maintained by PLAs that are in good and fair condition	16	16	Achieved

Justification of Overall Efficacy Rating

29. Based on description above, the efficacy is rated Substantial.

C. EFFICIENCY

30. At appraisal, an economic analysis was conducted for Component 1. The economic analysis of the flood and drainage management systems (Sub-components 1.1, 1.2 and 1.3) calculated avoided damages based on a with and without project scenario, while assuming a return period of fifty years using a simulated hydraulic model. The ex-ante analysis showed that these three sub-components, taken together as investments in the flood and drainage management system, were economically viable. Using a ten percent discount rate, the economic rate of return (ERR) was 11 percent, while the net present value (NPV) was LKR 971.8 million (US\$4.8 million, using the exchange rate at appraisal). Similarly, the analysis showed the economic viability for Component 1.4, as well as for Component 1 overall.
31. The ex-post analysis used an updated hydraulic model and also assumed a return period of fifty years, a ten percent discount rate, and the same methodology of avoided damages. The ERR for the base case is 16 percent, while the NPV is LKR 8,431 million; these are higher than the base case at appraisal, see Annex 4. The sensitivity analysis shows that the investment remains economically viable, even with increased costs. Further details of the economic analysis are provided in Annex 4.
32. Economic analysis was not carried out at appraisal for Component 2. Analysis for Component 2 used cost effectiveness to assess efficiency, as quantifiable data on benefits was not available to conduct a cost-benefit analysis. Investments under Component 2 were small and were designed to improve local infrastructure managed by the four PLAs. Annex 4 compares the cost per square meter for the two roads in Colombo under MCUDP with costs in Dhaka, Kandy, Karachi, Manila, and Nairobi. MCUDP costs are lower than in Dhaka, Manila, and Nairobi, and similar to the costs in Kandy, suggesting that these investments under MCUDP were cost effective. Costs in Karachi were however lower.
33. **Design efficiency.** The flood reduction system was designed on a framework basis at appraisal, which led to an increase in the time required for project completion. Several design phases were required for investments on flood reduction system, such as: feasibility studies, preliminary hydraulic studies, preliminary hydraulic designs. These were followed by the requirement on having detail design inclusive of structural design, geotechnical, electro-mechanical,



and instrumentation designs. The requirements for resettlements and land acquisitions could only be identified after the design phases. However, the rehabilitation of urban parks and small local infrastructure under the second component were ready for implementation and the project disbursed the allocated funds for Component 2 within four years of project implementation. The appraised implementation timeline of five years was too optimistic for a project of this scope and ambition. In addition, resettlement took significant time and led to implementation delays.

34. **Implementation efficiency.** At closure, the project had disbursed a total of US\$199 million, 93 percent of the total IBRD loan. The underutilization of the loan amount was in part due to the depreciation of the Sri Lankan Rupee towards the end of MCUDP, when a large final payment was made to contractors for the large Design and Build contracts of pumping stations and tunnels.
35. The extension of the original loan closing date by 48 months was due to the long procurement process for the four Design and Build contracts, which led to the project being in problem status for an extended period. Supervision and PMU costs also increased over the ten-year implementation period. At appraisal, design and build contracts were considered as the most appropriate option, with a single stage and two envelopes. However, during implementation, it took almost ten months to complete the procurement process. In 2017, only one of the four Design and Build contracts were under implementation, while others were still under procurement. All four contracts were under implementation only in 2019.

Efficiency rating: Modest

Justification of overall outcome rating

36. The overall outcome is rated **Moderately Satisfactory**, based on the above discussion and the ratings for the Relevance of the PDOs, Efficacy, and Efficiency.

D. OTHER OUTCOMES AND IMPACTS

37. **Gender.** The project ensured equal rights for men and women by capturing gender specific considerations in the Social Management Framework (SMF), which included: (i) allowing women as joint home owners with their spouses; (ii) livelihood restoration support for female-headed households; (iii) equal employment training opportunities for women; (iv) gender-sensitive provision of facilities at resettlement sites, including a center for women and daycare facilities; and (v) ensuring women's participation in the condominium management process. The project provided vulnerability allowances of LKR12.3 million for 56 female headed households living around macro-drainage interventions. At the resettlement site of Methsanda Sevana and Siyasettha Sevana, the building management block in charge of handling day-to-day issues included women at the management committee level. At Methsanda Sevana, four of the five key management positions are held by women, while at Siyasettha Sevana, four of the six managers are women. A beneficiary said that having women in building management helped them, as most day-to-day issues came from mothers and/or women who stay home and manage home-related issues.
38. **Institutional strengthening.** Prior to MCUDP, the Sri Lanka Land Development Corporation (SLLDC) had limited or no experience on urban tunneling and stormwater pumping stations. During the construction period, SLLDC, through the PMU, assigned designated staff for each intervention that was to be managed by SLLDC on their completion. Their capacity was built during the on-the-job training with contractors in geotechnical investigations, reviewing and



understanding preliminary and detailed designs, understanding the construction methodologies used in each intervention, design, and manufacturing, as well as installation and testing of electro-mechanical equipment. In addition, training was provided on project management and dispute resolution.

39. **Other unintended outcomes and impacts.** The project generated unintended positive impacts with respect to the improvement of wetlands. With the completion of the Beddagana wetland park, Colombo was accredited in 2018 as the first wetland capital city in the world by Ramsar International. The Department of Wildlife declared Kolonnawa Marsh, Kotte Marsh, and Heen Ela Marsh as Wildlife Sanctuaries in 2021. The Urban Development Authority included Dissayaru Park as the main wetland center in Colombo, and the Wetland Link International included Diyasaru Park and Beddagana Park as wetland members. Following the success of Beddagana Park as a biodiversity park, the Urban Development Authority launched the Diyasaru Wetland Kotte Rampart in October 2022 for educational tourism. Both wetland parks are open to the public and are managed by the Wetland Management Strategy Unit established under the project.

III. KEY FACTORS THAT AFFECTED IMPLEMENTATION AND OUTCOME

A. KEY FACTORS DURING PREPARATION

40. **The project was prepared as an urgent response to the two major floods that hit Colombo in 2010.** At that time, the capacity of the outfalls of the Colombo canal system was only for a return period of about ten-year storms in one day, assuming that all macro-drainage systems were well maintained. The project was designed urgently as a pioneering flood reduction system in the Colombo metro region to address floods along the Colombo basin. However, the time needed to build the entire system was underestimated. During the first year of implementation, the project started work on feasibility studies, running hydraulic models, and detailed designs for Component 1. In addition, it was challenging to select the best combination of interventions which would be feasible within the project period and the funds allocated under MCUDP.
41. **Implementation risk and the complexity of the institutional arrangements were inadequately assessed.** The project design was ambitious and most of Component 1 was designed on a framework basis, with the specific interventions and their locations not yet determined. Implementation risks, including the length of time to adjust for unexpected results from the hydraulic models or unidentified technical challenges, could have either been better assessed or contingency allowances could have been built in. The project had two implementation agencies for each component, and the resources needed for coordination and communication between these agencies to resolve implementation issues were underestimated. PMU staffing and the need for focal points from each agency could also have been more carefully estimated.
42. **Quick win activities under Component 2 were fully appraised.** At appraisal, the project had a clear plan for the implementation of Component 2. The high-priority infrastructure investments were identified upfront and implemented under the condition that the PLAs committed to budget for and undertake Operations and Maintenance (O&M) of the assets. Component 2 activities on capacity building and improvement in local infrastructure and urban public spaces were ready for implementation and were completed during the first four years of project implementation, as envisaged. Capacity building activities, as well as the purchase of equipment for PLAs to improve O&M, were supported by the project and were complemented by a parallel Trust Fund activity.



43. **The Results Framework was designed with a clear operational logic to achieve each target.** The indicators and targets for Component 1 were chosen based on the initial modelling conducted at the design stage and the PAD explicitly mentioned that some of them would need adjustments based on the hydraulic modelling, which was to be done during the first year of implementation. Component 2 indicators were focused on outputs which were easy to measure but were not entirely attributable to the improvement in the PLAs' capacity to operate and maintain these outputs.

B. KEY FACTORS DURING IMPLEMENTATION

Factors subject to the control of the World Bank

44. **Implementation challenges were mitigated at the first and second restructuring.** The project faced challenges during its initial years of implementation, as land acquisition, resettlement, and procurement processes required significant time. The project targets were reviewed at the MTR in December 2014. Interventions under Component 1 were revisited, and following the first restructuring in 2017, the project continued with priority investments with a realistic timeline to complete them by the revised closing date, along with the improvements to local infrastructure and urban open spaces.
45. **The Bank team identified institutional risks and introduced appropriate mitigation measures.** The risk of institutional coordination was addressed through dialog and regular interactions with the related agencies. PMU staffing challenges were tackled through strongly emphasizing the recruitment of additional staff and coordination across agencies. Fiduciary and safeguards issues were addressed through strict monitoring of procurement and safeguards processes. This included the provision of targeted training for PMU staff.

Factors subject to the control of the Government of Sri Lanka and implementation agency

46. The project experienced a high turnover within the main counterpart implementing agency. Throughout implementation, the project had at least four changes of Secretary¹⁰ of the Ministry. Therefore, several decisions were delayed despite efforts from PMU and the Bank to ensure smooth implementation. Shortcomings in coordination with the Colombo Municipal Council on resettlement meant that a small section of Norris Canal diversion could not be completed, see Social Safeguards section. However, the project maintained a solid Project Management Unit (PMU) with a highly competent Project Director who had extensive experience in engineering design and project management. Budget allocations were sufficient to maintain staffing throughout project implementation. Project and audit reports were submitted in a timely manner.

External factors beyond the control of the World Bank, GOSL, and the implementation agency

47. The Easter bombing in 2019 affected the implementation of ongoing sub-projects for two months due to security concerns and constraints, including curfews, movement restrictions, and temporary demobilization of some contractors. As some of the contractors and workers were fearful of the escalating violence, the project experienced difficulties in hiring and retaining construction personnel. Abnormally heavy rainfall, particularly in August and September 2019, which continued into December, resulted in the inundation of sub-projects. The unexpected COVID-19 pandemic impacted the ongoing operations of Chinese contractors, who faced labor shortages and significant

¹⁰ Secretary to the Ministry of Urban is the head of the Steering Committee and in charge from the implementation agency.



delays related to the manufacturing and shipment of critical equipment produced in China and Korea. All contractors submitted *force majeure* claims due to the COVID-19 pandemic, with a total delay of around 210 days. In 2021, less than one year before project closure, the country was faced with a major economic crisis, which led to the government starting to restrict the import of goods resulting in a major shortage of construction materials including cement, bitumen, and timber, which were crucial for the final stages of construction. The prices of materials increased more than 20 percent, and most contractors faced liquidity and cash flow issues.

IV. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISKS TO DEVELOPMENT OUTCOME

A. QUALITY OF MONITORING AND EVALUATION (M&E)

M&E Design

48. **The PDO and indicators for achieving a reduction in flood risk area were clear, but the second objective was more difficult to measure.** The target for PDO indicator #1 was assessed based on the results from the initial modeling on the reduction in risk area of floods from a 50-year return period. Intermediate indicators for Component 1 were appropriate to measure the achievements of the intermediate outcomes of the interventions on flood reduction. The Results Framework had shortcomings where Intermediate Results Indicators (IRI) were targeted more to investments and the achievement of outputs, rather than measuring the results of strengthened capacity to maintain these infrastructure investments. These shortcomings were subsequently addressed by introducing indicators on beneficiary satisfaction. Following the first restructuring, the PMU designed a beneficiary satisfaction survey which included indicators on the extent of satisfaction of the operations and maintenance of investments and assets under the responsibility of the four PLAs, as a proxy for the increase in the capacity of the PLAs.
49. **The project was designed to establish a Project Monitoring and Management System (PMMS) and Management Information System (MIS) for M&E.** This system was designed to be housed at the PMU. This includes improvement and the use of M&E system at the Ministry and SLLDC, as well as the M&E system in each participating municipal council. In addition, for the purpose of the Project, several features were added such as Procurement, Financial Management and Accounting, Safeguards, and technical progress. At appraisal, it was agreed that the project would provide capacity building support and support the purchase of equipment to improve the M&E of local infrastructure in each PLA.

M&E Implementation

50. As is typical urban projects, there were a few challenges in the implementation of M&E. It took several years until the MIS was developed for the project. A parallel activity financed by the Korea - World Bank Partnership Facility (KWPF) complemented Component 2 activities in developing and improving an urban Management Information System (MIS) and a Geographic Information System (GIS) mapping for Colombo Municipal Council and Sri-Jayawerdanapura-Kotte, and was initiated in 2018, after the first project restructuring. Implementation of M&E relied mostly on the manual database maintained by the PMU.
51. The COVID-19 pandemic affected the implementation of M&E and the updating of progress through the MIS. The last ISR, which was approved only a few days before the closing date, had incomplete information on the achievement of results indicators due to the challenges in collecting data.



M&E Utilization

52. **The Project Monitoring and Management Support (PMMS) was developed for MCUDP.** The PMMS was developed on the MCUDP website <https://pmms.mcudp.lk/administration/dashboard.php>, where information on the contracts for goods, works, and consultants, were monitored. A summary of PMMS collected by the PMU and utilized as the basis for discussion and field visits during the Bank's Implementation Support Missions. Following the completion of MCUDP, the SLLDC is using the Integrated Flood Management System (IFMS), which has a central SCADA system connected to the individual SCADAs installed in the three pumping stations and the flood gates constructed under the Project. Information and data on water levels and rainfall intensity from the IFMS are currently shared with the Department of Meteorology and other relevant agencies. A parallel MIS system, which was developed for CMC and Sri Jayawerdanapura-Kotte municipal councils, is being utilized as a tool for Operations and Maintenance (O&M) and for gathering information from various agencies to monitor assets and local infrastructure in these two local authorities.

Justification of Overall Rating of Quality of M&E

53. Based on the above discussion of M&E design, implementation and utilization, the overall rating of M&E quality is assessed as **Modest**.

B. ENVIRONMENTAL, SOCIAL, AND FIDUCIARY COMPLIANCE

Fiduciary compliance

54. **Procurement.** The project complied with the procurement procedures in the Loan Agreement and the Procurement Plan, and no waivers were sought. The quality of the bid documents was satisfactory in terms of both commercial requirements and technical specifications. Procurement documentation was in order throughout the project and no deviation from key fiduciary principles was reported. The project had no major fiduciary compliance issues during its implementation. The project experienced significant delays throughout implementation due to the prolonged time taken to finalize the designs, prepare tender documents and obtain approvals, technical evaluations, procurement committee approvals and contract awards. Implementation delays were experienced due to poor contractor performance, and some contractors were found to be deficient both in the allocation and mobilization of resources and in project planning. This was exacerbated during the final two years of implementation due to the COVID-19 pandemic.
55. **Financial Management.** During the initial stage of project implementation, the PMU had some issues related to FM staffing, which was subsequently addressed. Interim Unaudited Financial Reports (IUFRs) were submitted on time and were acceptable. Submission of annual external audit reports was also generally satisfactory, with a few delays of less than four months after the due date. Two audits were qualified with minor shortcomings, while the rest were unqualified. The project complied with the two main FM covenants in the Loan Agreement.

Safeguards compliance

56. Overall compliance with environmental and social safeguards was Moderately Satisfactory. The project was categorized as 'A' and the following safeguards policies were triggered: Environmental Assessment (OP /BP 4.01), Physical Cultural Resources (OP/BP 4.11), and Involuntary Resettlement (OP/BP 4.12), as well as the national policy



and legal framework for land acquisition, payment of compensation, and resettlements. The Environmental Assessment & Management Framework (EAMF) was prepared at appraisal and sub-project specific safeguard instruments based on the EAMF were prepared during implementation to address site-specific issues. Safeguards instruments were disclosed in the Bank website and in the government's project website. No significant, irreversible, and long-term impacts were recorded, as most of the project related impacts on the environment were short-term, localized, and construction related; these were managed moderately satisfactorily. No significant non-compliances on occupational health & safety (OHS) aspects were recorded during the project period, and the major environmental risks envisaged during the Talangama lake restoration, and the disposal of dredged desilting material were managed well by the PMU in consultation with relevant stakeholders. This was preceded by a basin-wide sediment analysis to determine the nature of the sediment waste as per the country's regulatory requirements on the disposal of hazardous waste.

57. Sub-projects implemented by MCUDP, especially those that involved canal diversions, canal bank improvements, and construction of new tunnels for the purpose of flood and drainage management in the Colombo metropolitan region required the acquisition of private land, recovery of state land occupied by squatter households, and the relocation and resettlement of project-affected households, largely the non-titleholder squatter families living along canal banks. Altogether, 652 households were affected by the project. Project alternatives were considered when introducing changes to the technical designs and minimizing land acquisition to avoid or minimize adverse resettlement impacts.
58. **Grievance Redress Mechanism (GRM).** The project engaged the affected parties as well as the other key stakeholders in a rigorous and continuous process of consultation and information sharing throughout the project cycle. This included consultations and awareness raising programs conducted at group and individual levels, and sharing printed material in local languages, containing project related information including the entitlements of affected parties and grievance redress procedures. The project also operated a four-tier grievance redress mechanism (GRM) across all its sub-projects which reported 407 grievances. Of these grievances, 93 percent (377) have been resolved, and the rest, largely related to compensation for construction related damages, will be resolved through the contractors via their respective insurance policies. The Bank will monitor the satisfactory resolution of these remaining grievances.
59. **Incomplete Resettlement at Norris Canal.** Relocation and resettlement of 24 households in a building belonging to the Colombo Municipal Council (CMC) was required for the completion of this sub-project. This building was structurally too weak to withstand the deep excavation works, and thus complete demolition was required. The 24 households comprised 16 tenants and 8 legal owners, who enjoyed unencumbered freehold interest for their housing units under a deed granted by the Colombo Municipal Council (CMC). To date, eight tenants are in the process of relocating, as the Supreme Court has decided that they should vacate the building. CMC's acquisition of the properties belonging to the four legal owners under Land Acquisition Act is in its early stages. At project closure, this sub-project was incomplete and GoSL agreed to finance the completion of the balance work of 126 meters. The project had diligently followed the resettlement process indicated in the RAP and there was no safeguards non-compliance. The Bank will continue to monitor the progress of the remaining resettlement.
60. Despite the non-completion of the 126 meters (of the total of 675 meters) of the Norris Canal diversion, the project successfully removed the bottleneck on the existing Norris Canal by constructing a new diversion of approximately 550 meters and connected lateral drains to the existing diversion channel. The improvement in drainage condition at



closure was evidenced during the heavy rains in Colombo in June 2022. Direct beneficiaries¹¹ indicated that there was only localized inundation, while the hospital at one end of the canal did not have any inundation. that.

C. BANK PERFORMANCE

Quality at Entry

61. The project was responding to an urgent need and high priority of the government. The team was responsive to the client's request and designed an integrated flood reduction system which was needed critically for Colombo area. However, more analysis on the feasibility of introducing large Design Build contracts in the project could have been conducted prior to implementation. The initial design of Component 1 was overly optimistic, but at the detailed design stage the Bank deployed sufficient experts, including hiring a highly experience hydrologist for Component 1. A municipal engineer was also part of the project design team. The project was also designed to improve PLA capacity to manage local infrastructures and activities under Component 2 were completed within the period of original closing date. However, the timeline for the implementation for Component 1, with its framework design involving multiple agencies, was underestimated.
62. The institutional capacity and ability of implementation agencies to resolve fiduciary and safeguards issues were overestimated. As discussed in the Social Safeguards section, the resettlement process was cumbersome and suffered from a lack of inter-agency coordination. The design underestimated the length of time for procurement processes. With several large contracts envisaged for the project at entry, this was the first such experience for the Ministry and the PMU, which led to implementation delays. The time needed prior to the commencement of construction was underestimated. Fiduciary, safeguards, and institutional arrangement risks were rated appropriately as substantial.

Quality of Supervision

63. **Proactive action to take the project out of problem status and turn it around as a transformational project.** The Bank team revisited the overall project at the MTR and addressed issues through the first restructuring in 2017. However, the project continued to remain in problem status, as most of the large design and build contracts for flood reduction were not awarded until 2018. Several resettlement issues were pending, and the procurement of large contracts was still in process. There was uncertainty on whether the project could achieve its objective without having the main contracts in place. Project implementation progressed significantly after all contractors had mobilized. The second restructuring included another extension of the loan closing date to ensure the completion of large contracts during the COVID-19 pandemic. The project rating was upgraded to Moderately Satisfactory following the second restructuring.
64. The project had four Bank Task Team Leaders (TTLs) and two co-TTLs, with all TTLs based in Colombo. This allowed the Bank to conduct close supervision, undertake regular check-ins with the PMU, and ensure that the PMU maintained sufficient staffing to implement the project.
65. The Bank fielded 21 implementation support missions during the project's nine years of implementation. The missions included a range of experts in urban development, civil engineering, urban management, environmental and social safeguards, procurement, and financial management. During the COVID-19 pandemic, the Bank team held

¹¹ Result from beneficiary satisfaction survey following project closure.



frequent video and audio conferences with the PMU. As all investments were in Colombo and the entire Bank team was based in Colombo, monthly check-ins and technical missions could be continued. Bi-weekly updates and monthly reporting by the PMU were strictly monitored by the Bank team to ensure project completion. The Aides Memoire were candid, covered the key issues, and provided practical guidance on resolving implementation issues. The efforts of the Bank safeguards and fiduciary specialists in ensuring compliance with Bank policies are summarized in the earlier section.

Justification of Overall Rating of Bank Performance

66. Based on the above discussion, Bank Performance is rated **Moderately Satisfactory**.

D. RISK TO DEVELOPMENT OUTCOME

67. **Institutional risk.** At closure, the Sri Lanka is facing a political and macro-economic crisis. There has been high staff turnover and changes in the government, which the risk that capacity built under the project is neither maintained nor mainstreamed either in the implementation agency or within various stakeholders responsible for operating and maintaining project-created assets. While flood reduction in the CMR continues to be a high priority for the government, as described in its development plan and the 2030 vision, institutional risk will remain substantial.
68. **Sustainability risk.** Sustainability of the infrastructure and services financed under MCUDP will depend on the SLLDC budgeting adequate resources to operate and maintain the completed flood reduction system under Component 1, and adequate O&M budget being allocated by the Colombo Municipal Council for micro-drainage and completed activities under Component 2. At closure, several steps have been taken to ensure dedicated budget for the newly constructed and rehabilitated infrastructure. At ICR stage, it was acknowledged that despite the ongoing crisis, the GoSL had allocated US\$2.4 million to SLLDC for O&M of the pumping stations and tunnels. This would need to be sustained during the life of the asset.

V. LESSONS AND RECOMMENDATIONS

69. **Accelerated project preparation in response to a situation of urgent need should recognize the longer timeline needed for the design and implementation of large construction contracts.** Innovative urban development and flood mitigation investments require adequate time for preparation, and options for the Bank lending instrument need to be carefully assessed. MCUDP was designed as a combination of a flood reduction system with large infrastructure, nature-based solutions, and urban upgrading in the heart of the most densely populated metropolitan area in the country. With the lack of prior experience and insufficient time for project preparation, the project was not ready for implementation on Board approval. Large infrastructure projects would need multiple agencies to implement and thorough analysis, which should include technical and institutional assessments. These interventions involve complex procurements and would also need close attention to environmental and social safeguards requirements. Future projects involving large construction works would need to allow sufficient time for preparation, including technical preparation, resettlement/land acquisition, and procurement processing.
70. **Exploring options for lending instruments for large and complex urban infrastructure projects.** At appraisal in 2011, the project assessed that the most suitable instrument available was an IPF of five years implementation. With the availability of the Multiphase Programmatic Approach (MPA), future projects should



consider an MPA, which would allow adequate time for preparation during Phase 1 of the MPA and focus on construction and sustainability in the subsequent phases.

71. **An integrated approach to flood reduction and urban upgrading should be combined with harmonized institutional arrangements.** While MCUDP achieved its expected outcomes at closure, there were challenges in institutional coordination throughout implementation. Smooth coordination was acknowledged between the Ministry of Urban Development and Housing and agencies within the purview of the Ministry, such as SLLDC and the Urban Development Authority (UDA). However, there were delays in implementation due to the lack of coordination between the implementation agency and the PLAs, such as the Colombo Municipal Council (CMC). This resulted in an incomplete resettlement and hence incomplete construction of the Norris canal diversion at project closure.
72. **Ensuring sustainability and functionality of completed interventions under the project requires discussions with and commitment from the agency responsible for operations and maintenance.** During its implementation, the project involved staff from SLLDC in the construction of macro-drainage interventions, as SLLDC will be the asset owner after completion. Transfer of knowledge and on the job training for the operations and maintenance of these interventions were necessary for their sustainability. At closure, SLLDC staff were trained, manuals for the completed constructions were translated in the local languages, and the government provided sufficient budget to the Ministry of Urban Development and Housing and SLLDC for staffing and budgeting of the completed interventions, to ensure their functionality. During the high rainfall in June 2022, the pumping stations were utilized and functioned effectively. As the country is currently in a financial crisis with limited budgetary resources, O&M of urban parks and wetlands, which is the responsibility of the Urban Development Authority and the municipal councils, has been neglected. For these types of interventions, future projects could consider joint O&M with the private sector, potentially outsourcing maintenance to third parties.



ANNEX 1. RESULTS FRAMEWORK AND KEY OUTPUTS

A. RESULTS INDICATORS

A.1 PDO Indicators

Objective/Outcome: Reduce flooding in the catchment of the Colombo water basin

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Reduction in the area under risk of flooding (50-year return period) in the project area.	Square kilometer(km2)	0.00	2.00	3.00	3.03
		16-Jul-2012	16-Jul-2012	18-Sep-2017	31-Dec-2021

Comments (achievements against targets):

On June 10, 2016, the Project Management Unit provided an update of the impact on the PDO if Madiwela Diversion is not completed. The end target was revised as part of the restructuring in 2017. The revised end target was based on technical analysis to better align with the indicator and likelihood to achieve the target by project closing date.

Objective/Outcome: Strengthen capacity of local authorities

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Percentage of total urban	Percentage	50.00	70.00		70.00



roads maintained by PLAs that are in good and fair condition in the 4 PLAs		16-Jul-2012	16-Jul-2012	18-Sep-2017	31-Dec-2021
Comments (achievements against targets):					

A.2 Intermediate Results Indicators

Component: Component 1: Flood and Drainage Management

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
1.1 Length of primary canals improved by the project	Kilometers	0.00	9.20	9.20	10.62
		16-Jul-2012	16-Jul-2012	16-Jul-2012	31-Dec-2021
Comments (achievements against targets):					
All interventions for improved primary canals were fully completed and exceeded the indicator target.					

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
1.2 Increase in drainage capacity (gravity system)	Cubic Meter(m3)	100.00	185.00	185.00	220.00
		16-Jul-2012	16-Jul-2012	16-Jul-2012	31-Dec-2021



Comments (achievements against targets):

Drainage capacity was improved following completion of macro-drainage interventions and it exceeded the target.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
1.3 Increase in drainage capacity (pumping system, subject to the results of the hydrological model)	Cubic Meter(m3)	3.50	25.00	25.00	60.00
		16-Jul-2012	16-Jul-2012	16-Jul-2012	31-Dec-2021

Comments (achievements against targets):

Fully achieved by completion of North Lock pumping station, St. Sebastian South, and Ambatale pumping station. The total capacity of these 3 pumping stations are 60 m3.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
1.4 Micro-drainage subprojects implemented under the Project that reduce risk of flooding in localised areas (10 year return period)	Number	0.00	7.00	6.00	5.00
		16-Jul-2012	16-Jul-2012	22-Jun-2020	31-Dec-2021



Comments (achievements against targets):

At appraisal, 15 micro drainage was proposed by the municipal council. At implementation, the project focused on 6 micro drainage improvements based on their feasibility. At closure, the project completed 5 and partially completed Norris Canal diversion.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
1.6 User's satisfaction with the condition of Beira Lake and Beddagana and Rampart Park	Percentage	31.00	80.00	80.00	83.26
		30-Apr-2014	16-Jul-2012	16-Jul-2012	31-Dec-2021

Comments (achievements against targets):

Beneficiary satisfaction survey was conducted for Beddagana Wetland Park, Viharamadevi Park, and Kotte Rampart. Respondents were direct beneficiaries, of which 83 percent were satisfied with the completed interventions under the project.

Component: Component 2: Urban development, infrastructure rehabilitation and capacity building for Metro Colombo local authorities.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
2.2 Km of road/drainage built/rehabilitated based on prescribed standards	Kilometers	10.00	45.00	45.00	47.00
		16-Jul-2012	16-Jul-2012	16-Jul-2012	31-Dec-2021

Comments (achievements against targets):



Major road around Colombo Metro Area were rehabilitated. In total, 47 km road rehabilitated including sidewalks. This exceeded the target and all were completed within the first 4 years of project implementation.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
2.6 Users satisfied with area-based demonstration projects in Colombo City	Percentage	0.00	70.00	70.00	78.21
		30-Apr-2014	16-Jul-2012	16-Jul-2012	31-Dec-2021

Comments (achievements against targets):

Direct beneficiaries indicated satisfaction of improved public parks including Linear Park, Townhall area, and Viharamadevi Park.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
2.7 Number of public spaces / parks upgraded and revitalized by the project	Number	0.00	3.00	3.00	3.00
		30-Apr-2014	16-Jul-2012	16-Jul-2012	31-Dec-2021

Comments (achievements against targets):

3 major public parks in the heart of Colombo were successfully improved and maintained by the municipal council and the urban development authority.



Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
2.8 Public convenience complexes maintained by PLAs that are in good and fair condition.	Number	0.00 30-Apr-2014	16.00 16-Jul-2012	16.00 16-Jul-2012	16.00 31-Dec-2021
Comments (achievements against targets): 16 public toilets around Colombo, including within public parks and wetland parks were improved and maintained by the municipal council.					
Component: Component 3: Implementation Support					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
3.1 RTC developed and integrated into SLLRDC's operation	Yes/No	No 16-Jul-2012	Yes 16-Jul-2012	Yes 16-Jul-2012	Yes 31-Dec-2021
Comments (achievements against targets): Real Time Center developed under the project, located in a building constructed using GOSL funds under the project. The IFMS is housed in this building and managed by SLLDC and integrated to its system.					

B. KEY OUTPUTS BY COMPONENT



Objective/Outcome 1 Reduce flooding in the catchment of the Colombo water basin	
Outcome Indicators	Reduction in the area under risk of flooding (50-year return period) in the project area.
Intermediate Results Indicators	1. Length of primary canals improved by the project 2. Increase in drainage capacity (gravity system) 3. Increase in drainage capacity (pumping system) 4. Micro-drainage subprojects implemented under the Project that reduce risk of flooding in localised areas (10 year return period) 5. User's satisfaction with Beira Lake ,Beddagana and Rampart Park
Key Outputs by Component (linked to the achievement of the Objective/Outcome 1)	1. 60 m³ pumping stations improved 2. 2 tunnels at New Mutwal and Torrington were developed 3. 6 micro-drainages in Colombo Metro area were improved 4. 220 m³ drainage system were improved
Objective/Outcome 2 Urban development, infrastructure rehabilitation and capacity building for Metro Colombo local authorities	
Outcome Indicators	% of total urban roads maintained by PLAs that are in good and fair condition in the 4 PLAs
Intermediate Results Indicators	1. Km of road/drainage built/rehabilitated based on prescribed standards 2. Users satisfied with area-based demonstration projects in Colombo City 3. Number of public spaces / parks upgraded and revitalized by the project 4. Public convenience complexes maintained by PLAs that are in good and fair condition.
Key Outputs by Component (linked to the achievement of the Objective/Outcome 2)	1. 47 km of roads and sidewalks rehabilitated 2. 3 public parks improved 3. 16 public conveniences improved

**ANNEX 2. BANK LENDING AND IMPLEMENTATION SUPPORT/SUPERVISION****A. TASK TEAM MEMBERS**

Name	Role
Preparation	
Rosanna Nitti	Task Team Leader(s)
Haider Raza	Procurement Specialist(s)
Supul Chamikara Wijesinghe	Financial Management Specialist
Michelle Lisa Chen	Team Member
Nadeera Rajapakse	Team Member
Satya N. Mishra	Social Specialist
G. W. Anjali U. Perera Vitharanage	Team Member
Zhiyu Chen	Team Member
Darshani De Silva	Team Member
Priyantha Jayasuriya Arachchi	Team Member
Niluka Nirmalie Karunaratne Sriskanthan	Team Member
Supervision/ICR	
Thalyta Ernandya Yuwono	Task Team Leader(s)
G. W. Anjali U. Perera Vitharanage	Procurement Specialist(s)
Bernadeen Enoka Wijegunawardene	Financial Management Specialist
Laksiri Priyadarshana Chamindr Weerackody	Team Member
Zeenath Marikar	Procurement Team
Samantha Prasada Wijesundera	Team Member
Roderick M. Babijes	Team Member
Kristine Schwebach	Social Specialist
Shakil Ahmed Ferdausi	Environmental Specialist

**B. STAFF TIME AND COST**

Stage of Project Cycle	Staff Time and Cost	
	No. of staff weeks	US\$ (including travel and consultant costs)
Preparation		
FY11	38.879	228,495.20
FY12	72.820	423,021.15
Total	111.70	651,516.35
Supervision/ICR		
FY12	15.986	138,234.76
FY13	82.205	658,492.90
FY14	58.638	1,069,800.04
FY15	22.780	414,576.08
FY16	34.130	271,638.46
FY17	26.994	242,829.54
FY18	37.226	199,075.47
FY19	46.129	221,149.37
FY20	38.778	276,273.73
FY21	37.138	226,032.73
FY22	24.223	171,097.82
FY23	1.272	5,117.43
Total	425.50	3,894,318.33

**ANNEX 3. PROJECT COST BY COMPONENT**

Components	Amount at Approval (US\$M)	Actual at Project Closing (US\$M)
Component 1: Flood and Drainage Management	147.55	149.66
Component 2: Urban development, infrastructure rehabilitation and capacity building for Metro Colombo local authorities.	50.70	48.30
Component 3: Implementation Support	4.01	1.10
Contingency and Front end fee	19.84	
Total	213.00	199.06¹² (98.4%)

¹² Due to exchange rate depreciation, the total amount disbursed was smaller than the number proposed at preparation.

**ANNEX 4. EFFICIENCY ANALYSIS****Economic Analysis at Appraisal**

1. The analysis for components 1.1-1.3 used a with and without project scenario. The project scenario is predicted to reduce the inundated area and therefore damages from floods. A hydraulic model, based on a 2002 study, was used to estimate flood frequency and associated damages. Benefits were estimated using the avoided damages method based on these modelling estimates for different flood frequencies. The flow of costs and benefits were measured for the lifetime of the assets – estimated at fifty years at the maximum. The results of the cost-benefit analysis (CBA) for Components 1.1-1.3 suggested the project was economically viable. Using a ten percent discount rate, the ERR was 11 percent while the NPV was LKR 971.8 million. The sensitivity analysis showed that the ERR was robust to different scenarios, including a 10 percent increase in costs, and a 10 percent decrease in benefits. The analysis at appraisal also included estimates for component 1.4 (based on projected increases in land value) and Component 1 in the aggregate. The results for the economic analysis at appraisal for Component 1 are shown in table A4.1 below.

Table A4.1: Results of Cost Benefit Analysis at Project Appraisal

	ERR	NPV (LKR million)¹³
Flood and Drainage Management (Sub-components 1.1, 1.2 and 1.3)	11%	971.8
Beira Lake Development (Sub-component 1.4)	28%	2,067.9
Flood and Drainage Management (Component 1)	12%	2,949.3

Economic Analysis at Completion

2. The economic analysis at completion is focused on a CBA of Components 1.1 – 1.3 and an assessment of cost effectiveness for Component 2. As data on benefits of Component 1.4 were not available or quantifiable, this was not included.

Cost Benefit Analysis for Components 1.1 – 1.3.

3. The economic analysis at completion tried to follow the methodology used at appraisal, with updated data as possible and available. Costs were based on actual expenditures. There are numerous benefits from the investments, however, the ones that could be quantified and used in the analysis comprised of two main types: (i) direct benefits, in the form of avoided damages from reduced flooding; and (ii) indirect benefits, captured through an increase in the value of land. The economic costs and benefits are considered to be 95 percent of the financial costs and benefits in the analysis. This is to reflect market factors such as taxes and duties.

Direct Benefits

4. Direct benefits were quantified by using an updated simulation of flooding in the Colombo Metropolitan area. Whereas the analysis at appraisal used data from 2002 and 2010, this hydraulic model uses updated rainfall data to estimate rainfall and subsequent inundation, with the improved drainage system. It assumes a return period of fifty years. It is estimated that there will be a reduction of 3.03 km²

¹³ For comparison purposes, these have been converted to LKR from USD using the exchange rate for 2011.



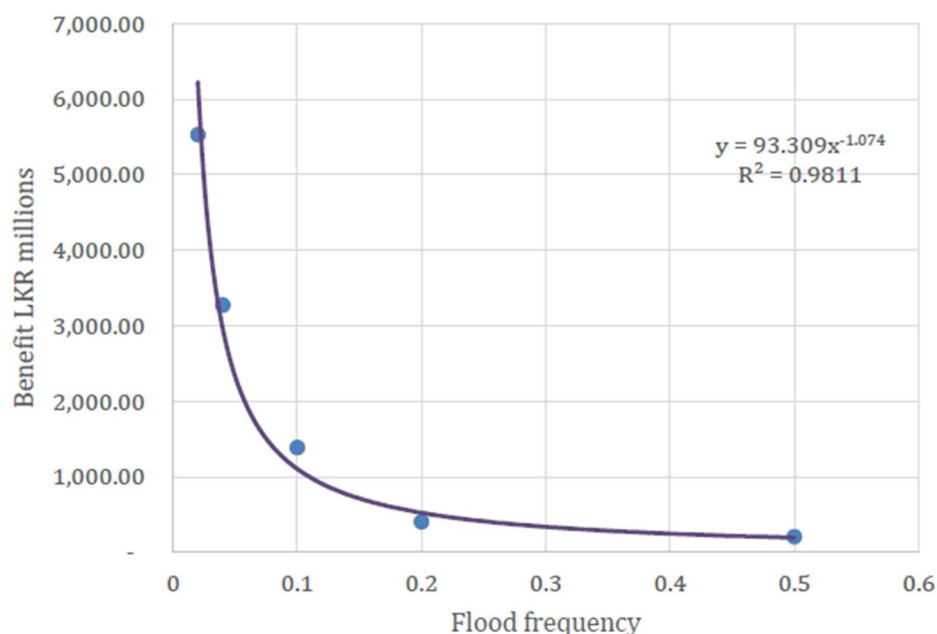
in the area inundated – the people in this area, estimated at 29,700 using 2012 census data, directly benefit due to the damages avoided from flooding.

5. Using the reduction in the inundated area, we can determine the avoided damages (i.e., the benefits). We used data from the Damage Assessment Methodology Report for the floods in 2016 to estimate damages by type of building. There are two types of damages – (i) structural damage to the building, and (ii) content damage, which is damage to contents of the building. The reduction in structural damage is estimated to be LKR 3,807.94 million, while for the content damage it is estimated as LKR 1,721.27 million. The reduction in total damage is therefore LKR 5,529 million for a 50-year return period. Using a similar methodology, we estimate the damages avoided (benefits) for several different return periods, as shown in table A4.2 below. This data (flood probabilities) is then plotted using the frequencies mentioned in the table and fitted to a power series, as shown in Figure A4.1. Integration of the obtained function gives the average annual avoided damages for 2021, which is equivalent to LKR 605.31 million.

Table A4.2: Avoided damages for various return periods

Return Period (years)	Frequency	Avoided Damages (LKR millions)
2	0.5	206.85
5	0.2	404.43
10	0.1	1,387.47
25	0.04	3,274.63
50	0.02	5,529.00

Figure A4.1. Probability distribution function for avoided damages



6. However, the avoided structural and content damages are not equal to the recovery needs, which is the actual total benefit of the investments. These are calculated using the Post-Disaster Needs Assessment Report 2016, which estimates it by sector for the disaster in 2016. Using damages estimated



for housing and the data from this report, we can use proportions – assuming similar proportions in our model – to estimate damages and losses for other sectors (e.g., food, livestock, transport, health, education, energy, water and sanitation, and so on). This method is used to obtain the expected annual average damage, which yields a figure of LKR1,715.82 million for 2016. Updating this for inflation to 2021, we anticipate an expected annual benefit of LKR 2,219.8 million – this is the total direct benefits due to the avoided damages and losses from the investments. The benefits are assumed to kick in for the project after completion – as the project finished in December 2021, the analysis assumes the benefits accrue from 2022.

Indirect Benefits

7. There are several indirect benefits, which are not tangible, such as an improvement in the overall livability of the city and increased recreational value of the canal system. However, we capture some of these in the form of a land value increase – assuming that the improved drainage and reduced flooding will enhance the value of the land, particularly in areas that would otherwise have been inundated. We capture this increase as a one-time benefit in our CBA. This is because, given the large area under consideration and dynamic environment, land value increases are not fully attributable to the project itself necessarily, so we cannot isolate and use this as a continuous benefit in our analysis.

8. The land value in a without project scenario is obtained from the actual cost of land acquisition under the project in 2016 – this is the base price. Prices in 2022 were then determined through a survey carried out by the PMU. The Central Bank of Sri Lanka publishes a Land Value Index (LVI) for Colombo – we use the LVI to project the base prices from 2016 to 2022. This would be the price without the project in 2022. We then subtract these by the prices obtained in the survey. This value gives us the benefit – however, given uncertainties in the survey as well as Sri Lanka’s macroeconomy, we only take 80 percent of these values as the benefit to keep our estimate conservative. It should be noted that these values are only taken for those areas which do not flood now, and only for land that is used for residential, commercial, and industrial uses.

Operation and Maintenance Costs

9. The operation and maintenance costs were estimated in detail by engineers of the PMU (see table A4.3). These costs are incurred after project completion in 2022. In line with the PAD, and assuming Sri Lanka’s macroeconomic outlook stabilizes, we assume that both, the operation and maintenance costs and the benefits, increase each subsequent year by 5 percent. In addition, a one-time cost incurred after 20 years (in 2042) is included to account for major replacement costs of mechanical parts.

Table A4.3: Estimation of the operation and maintenance costs for components 1.1 – 1.3 in 2022 LKR.

	Monthly Cost	Quantity	Annual Cost
South lock pumping station			
Pump maintenance	41,667	1	500,000
Gate maintenance	208,333	1	2,500,000
Trash screen maintenance	41,667	1	500,000
Office operation and maintenance	7,500	1	90,000
Engineering Assistant staff cost	136,272	2	3,270,528
Operator staff cost	130,958	0.1	157,150
Helper cost	111,271	2	2,670,509
Annual material and staff			9,688,187



Annual power cost assuming 5 pumping months with 24 hours of pumping per month	2,113,920	5	10,569,600
Total			20,257,787
Ambatale pumping station			
Pump maintenance	41,667	1	500,000
Gate maintenance	208,333	1	2,500,000
Trash screen maintenance	41,667	1	500,000
Office operation and maintenance	7,500	1	90,000
Engineering Assistant staff cost	136,272	2	3,270,528
Operator staff cost	130,958	0.1	157,150
Helper cost	111,271	2	2,670,509
Annual material and staff			9,688,187
Annual power cost assuming 5 pumping months with 24 hours of pumping per month	3,420,000	5	17,100,000
Total			26,788,187
North lock pumping station			
Pump maintenance	41,667	1	500,000
Gate maintenance	208,333	1	2,500,000
Trash screen maintenance	41,667	1	500,000
Office operation and maintenance	7,500	1	90,000
Engineering Assistant staff cost	136,272	2	3,270,528
Operator staff cost	130,958	0.1	157,150
Helper cost	111,271	2	2,670,509
Annual material and staff			9,688,187
Annual power cost assuming 5 pumping months with 24 hours of pumping per month	4,104,000	5	20,520,000
Total			30,208,187
Gate Stations			
Cost for operating Flushing gates	83,333	3	3,000,000
KCD 1 gate operations	83,333	1	1,000,000
KCD 4 gate operations	83,333	1	1,000,000
Thalangama gate operations	83,333	1	1,000,000
Total			6,000,000
RTC cost			
Staff cost (8 persons)	1,012,153	1	12,145,839
Internet, Cloud and other maintenance	800,000	1	9,600,000
Total			105,000,000

Final Economic Analysis

10. The final economic analysis results at completion, using the costs and benefits described above, are shown in table A4.4. In the base case, using a ten percent discount rate the NPV is LKR 8,389 million while the ERR is 16 percent, suggesting the project is economically viable. This is also higher than the results of the analysis at appraisal. Furthermore, the sensitivity analysis shows that the project remains viable even under decreased benefits and increased costs. With a 20 percent decrease in benefits (sensitivity case 1), the NPV is LKR 4,154 million and the ERR is 13 percent. Under even more conservative



estimates, in the sensitivity case 2 the project remains viable with a 20 percent increase in costs and a 20 percent decrease in benefits – the NPV here is LKR1,598 million and the ERR is 11 percent. The analysis is therefore robust to a range of different scenarios.

Table A4.4: Economic analysis for components 1.1 – 1.3 at completion

	Base Case (million LKR)	Sensitivity Case 1 - 20% decrease in benefits	Sensitivity Case 2 - 20% increase costs and 20% decrease in benefits
Economic Rate of Return (%)	16%	13%	11%
NPV (LKR Million)	8,389	4,154	1,598

Cost Effectiveness for Component 2

11. As data on benefits for Component 2 was not available or quantifiable, we analyze this component for cost effectiveness. In particular, we compare costs per square meter for road construction under MCUDP to that in other cities (Dhaka, Kandy, Karachi, Manila and Nairobi) For MCUDP, the costs were estimated for two road projects under component 2 – Galle Road A and B. In 2021, their costs per square meter were estimated as LKR 3,880 and 4,686 respectively.

12. The costs used here were for the surface applications, with a surface thickness of 100 mm. For Galle Road A the wearing/binder proportion was 40 and 60 percent, while for Galle Road B this was 50 percent each. The costs obtained for each of the other comparator cities used either one of these proportions to keep the comparison similar. In addition, as prices obtained were for various years and in different currencies, they were all first converted to 2021 prices in their local currency and then converted to LKR using the average exchange rate for 2021. This enables us to compare them. The costs per square meter for MCUDP funded roads are lower than costs for a similar road in Dhaka, Manila, and Nairobi. The cost per square meter in Kandy is between the cost for Galle Road A and Galle Road B, suggesting that there is a range of costs in Sri Lanka. Finally, only costs in Karachi are lower than that in Colombo. Overall, the projects under MCUDP are relatively cost effective when compared to metro areas in other countries.

Table A4.5: Comparing cost per square meter for road construction in various cities for 2021

	Dhaka ¹⁴	Kandy ¹⁵	Karachi ¹⁶	Manila ¹⁷	Nairobi ¹⁸	Galle Road A	Galle Road B
Cost m² LKR 2021	6,569	4,104	2,720	10,133	8,033	3,880	4,686

¹⁴ Source: "RHD Schedule of Rates 2019," Roads and Highways Department, Ministry of Road Transport and Bridges, Government of Bangladesh.

¹⁵ Data obtained from PMU of World Bank funded Sustainable Cities Development Project.

¹⁶ Data obtained from Karachi Municipal Corporation.

¹⁷ Source: JICA Report (https://openjicareport.jica.go.jp/pdf/12113353_02.pdf)

¹⁸ Source: "Cost Estimation Manual for Road Maintenance Works 2019," Ministry of Transport, Infrastructure, Housing, Urban Development and Public Works, Government of Kenya.



ANNEX 5. BORROWER, CO-FINANCIER AND OTHER PARTNER/STAKEHOLDER COMMENTS

We would like to thank the World Bank for supporting the Government of Sri Lanka to implement Metro Colombo Urban Development Project (MCUDP) to relieve the capital of Sri Lanka from flooding and to improve infrastructure facilities and urban livelihood. Below are the comments (chapter-wise) to the Bank ICR, given by the Ministry of Urban Development and Housing as the executing agency.

I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES

We are happy to note that MCUDP was aligned with the WBG's Sri Lanka Country Partnership Framework (CPF) FY 17-FY20 and PDO outcomes were fully aligned with the CPF's Area Engagement 3: Green Growth and Resilience and Objective 3.1. Improving urban livability, to develop urban areas in a more environmentally friendly manner.

II. OUTCOME

Efficacy

It is a pleasure that a considerable majority (81.49%) of beneficiaries were found to be confident and in good impression that flood risk has been declined as a result of interventions under MCUDP, in the MCUDP Beneficiary Survey conducted by the World Bank. **Table 4 - Indicators for objective 2:** MCUDP received the expert support from University of Moratuwa in evaluating road roughness. They conducted an Evaluation of International Roughness Index (IRI) Value of Urban Roads Maintained by PLAs. Roads with an IRI below 7 are considered to be in good condition, therefore 94% of the roads are in good condition even after 5 years after completion exceeding the target. *We would like to thank World Bank team for identifying the need of an IFMS (Integrated Flood Management System) at the time of appraisal (10 years ago) and including it under Flood and Drainage Management component. The system has been very useful in managing the recent flood events and will be more useful in the context of climate change.*

Efficiency

Design Efficiency: The component 1, Flood and drainage management was designed as a framework at appraisal. A better understanding of the hydrological aspects was required to determine the interventions under component 1. The interventions were finalized after conducting hydrological studies and modelling. The process included feasibility studies, preliminary hydraulic studies and designs followed by detailed designs inclusive of structural, geotechnical, electromechanical and instrumentation designs. Though it was time consuming, completion of these studies led to identification of most suitable combination of solutions for the flooding issue, which could be implemented within the available funds. The interventions under the component 1 were finalized at the Mid-Term review conducted in November 2014. Further, land acquisition and resettlement needs could only be identified after finalizing the interventions. As a result, land acquisition and resettlement have been carried out during the project duration itself, rather than as pre-project activities.

Implementation Efficiency: It is a known fact the MCUDP had to face two force-majeure situations during its implementation. Easter Sunday bomb attack and prolonged effects of the Covid-19 pandemic hindered the rate of progress of implementation in this urban context. As a result, MCUDP took a longer period for implementation. It was a huge challenge to continue construction activities during the Covid-19 pandemic, mainly due to country lockdowns and disconnections in the global supply chains.



We would like to express our sincere gratitude to the World Bank colleagues for supporting us throughout the two and half years of Covid-19 impact period to successfully complete a substantial amount of work amidst all the challenges even without compensating for the loss of time (210 days) due to the pandemic. For better planning and implementation, our opinion is that the project should have been structured in two phases: Phase one for feasibility, identification of interventions, land acquisition, resettlement and environmental clearances (pre-project phase) whereas phase two for procurement and implementation (implementation phase).

VI. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISK TO DEVELOPMENT OUTCOME

Safeguards compliance

The project operated a web-based four-tier grievance redress mechanism (GRM) across all its subprojects which altogether reported 423 grievances. Being a web-based mechanism, affected parties could submit a grievance through the online system available at

<https://social.mcudp.lk/administration/dashboard.php>.

This facilitated both the Bank and the borrower to check the status of a grievance, ensuring the transparency of the entire GRM. This GRM was featured and appreciated at several World Bank forums.

Bank performance

We wish to express our gratitude once again to the World Bank team for taking proactive actions to take the project out of problematic status and turn it around as a transformational project.

At the time of second restructuring (June 2020), Covid-19 pandemic was having a continuous impact. Understanding of the Bank during the mission was that the impact from Covid-19 will be enormous, and their instinct was, it is too early to determine the true impact of the pandemic, which was seen as an occurrence that must be compensated in the future.

We are extremely grateful to the World Bank team for closely overlooking the project and supporting its implementation functions throughout the entire project period until the very end ensuring its successful completion. The project implementation agency staff received substantial exposure and knowledge transfer, which resulted in a lot of capacity development opportunities in technical skills in diversified areas of expertise. The Bank's emphasis on social and environmental safeguard compliance ensured the development activities caused maximization of positive impacts and minimization and mitigation of negative impacts to the public and to the environment. The Bank also facilitated the project staff with opportunity to work together with the Bank officials and international experts, which formed a capable team from PIAs to bear the responsibility of delivering future development endeavors.

Submitted on behalf of the Ministry of Urban Development and Housing

DG (Development), Ministry of Urban Development & Housing -Eng (Mrs.), Eng. Devika Abeysuriya



ANNEX 6. SUPPORTING DOCUMENTS

Document Name	Document Number
Aide Memoire – Project Launch & First ISM Mission – May 21-June 1, 2012 ISR – December 27, 2012	ISR #1
Aide Memoire – Third ISM Mission – Jan – April 2013 ISR – June 22, 2013	ISR #2
Aide Memoire – Fourth ISM Mission – Oct 7-15, 2013 ISR – December 13, 2013	ISR #3
Aide Memoire – ISM Mission – May 5-13 & May 26-30, 2014 ISR – June 26, 2014	ISR #4
Aide Memoire – MTR Mission – Nov 24-Dec 5, 2014 ISR – December 4, 2014	ISR #5
Aide Memoire – ISM Mission – May 11-17 & June 10-13, 2015 ISR – June 19, 2015	ISR #6
Aide Memoire – ISM Mission – Sept 14-21, 2015 ISR – September 12, 2015	ISR #7
Aide Memoire – Follow-up Visit to Supervision – Dec 7-11, 2015	
Aide Memoire – Follow-up Visit to Supervision – April 25-May 4, 2016 ISR – April 18, 2016	ISR #8
Aide Memoire – ISM Mission – Nov 13-18, 2016 ISR – July 11, 2016	ISR #9
Aide Memoire – ISM Mission – April 28-May 5, 2017 ISR – February 17, 2017 ISR – August 23, 2017	ISR #10 ISR #11
Aide Memoire – ISM Mission – Jan 29-Feb 8, 2018 ISR – March 23, 2018	ISR #12
Aide Memoire – Project Update Aug 2018 (PP Presentation) ISR – September 27, 2018	ISR #13
Aide Memoire – ISM Mission – April 1-9, 2019 ISR – April 10, 2019	ISR #14
Aide Memoire – ISM Mission – Aug 22 – Oct 30 2019 ISR – October 2019	ISR #15
ISR – December 2019	ISR #16
Aide Memoire – ISM Mission – Feb 17 – March 13, 2020 ISR – May 2, 2020	ISR #17
Aide Memoire – ISM Mission – Nov 2-16, 2020 ISR – December 15, 2020	ISR #18
Aide Memoire – ISM Mission – Feb 16 – March 2, 2021 ISR – March 30, 2021	ISR #19
Aide Memoire – ML Only – Aug 17 Sept 15, 2021 ISR – September 21, 2021	ISR #20
Aide Memoire – ISM Mission – Nov 29 – Dec 14, 2021 ISR – December 26, 2021	ISR #21
Restructuring Paper – August 2017	RES22390
Restructuring Paper June 2020	RES 30599
Beneficiary satisfaction survey for Viharamadevi Park, Beddagana Park, and Beach Crow Park - 2020	
Beddagana Wetland Report - 2018	
Beneficiary Satisfaction Survey for Component 1 and Component 2 after project closure – 2022	

Additional supporting documents

Parliament Area – 2010 Flood event



Lock Gate and Pumping Station





Thunmulla Junction, one of the busiest intersections in the middle of Colombo city. Rainfall event, May 2016 (288 mm/day)



Tunnels Network – New Mutwal and Torrington





Integrated Flood Management System – www.pub.curwsl.org

Monitoring weather, water level, and rainfall intensity

Information available for: Sri Lanka Land and Development Corporation (SLLDC), Colombo Municipal Council (CMC), Disaster Management Center (DMC), and Local Authorities within DSD in Colombo District



• Situation details

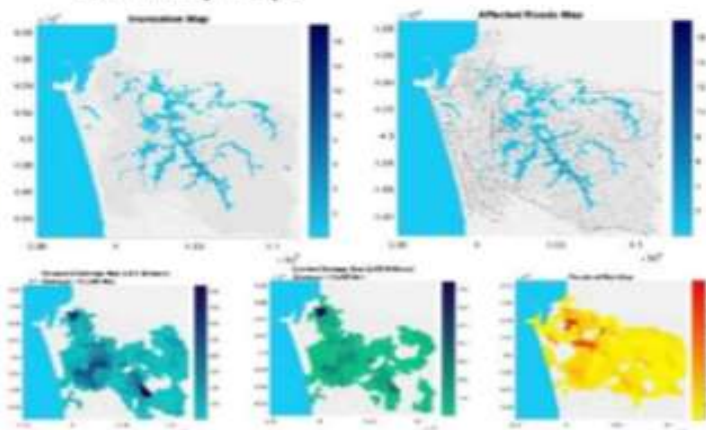
Rainfall

Rainfall gauge	Past hour	Past 3 hours	Past 12 hours	Past 24 hours
Melabe	17	58	45	55
Kaduwela	10	80	60	75
Battaramulla	00	20	45	50
Betapotha	8	10	15	20
Thalawathigoda	5	15	50	60
Hokandara East	08	32	45	45

Water level

Rainfall gauge	Past hour	Past 3 hours	Past 12 hours	Past 24 hours
Thalangama Tank	1.282	1.304	1.352	1.325
Amariyana Tank	0.986	0.875	0.861	0.874
Chandrika	0.981	0.946	0.952	0.953
Komarasinga Mv				
Low level bridge- Kaduwela	2.821	2.756	2.742	2.751
Kaduwela bridge	3.000	3.000	3.111	3.165

• Summary maps



- Expected number of people at risk (affected by flood levels > 0.3 m, map prepared for > 50 people per GND)

DSD	GND	GND		GND	GND
		WTD	WTD		
COLPBO	41,952	280	COLPBO	40,000	2024
BETHALA	2,951	11	COLPBO	40,000	0
KODAMELA	23,841	144	COLPBO	40,000	5,000
KOLTHANA	50,247	117	COLPBO	40,000	10,000
KARUNAKA	730	101	COLPBO	40,000	0
SRI JAYAWARDENAPURA KOTTE	8,655	120	COLPBO	40,000	1,000
THORAKKALUWA	11,288	10	COLPBO	40,000	1,000

Colombo Municipal Council Townhall Area and Viharamadevi Park – 50.8 hectares improvement

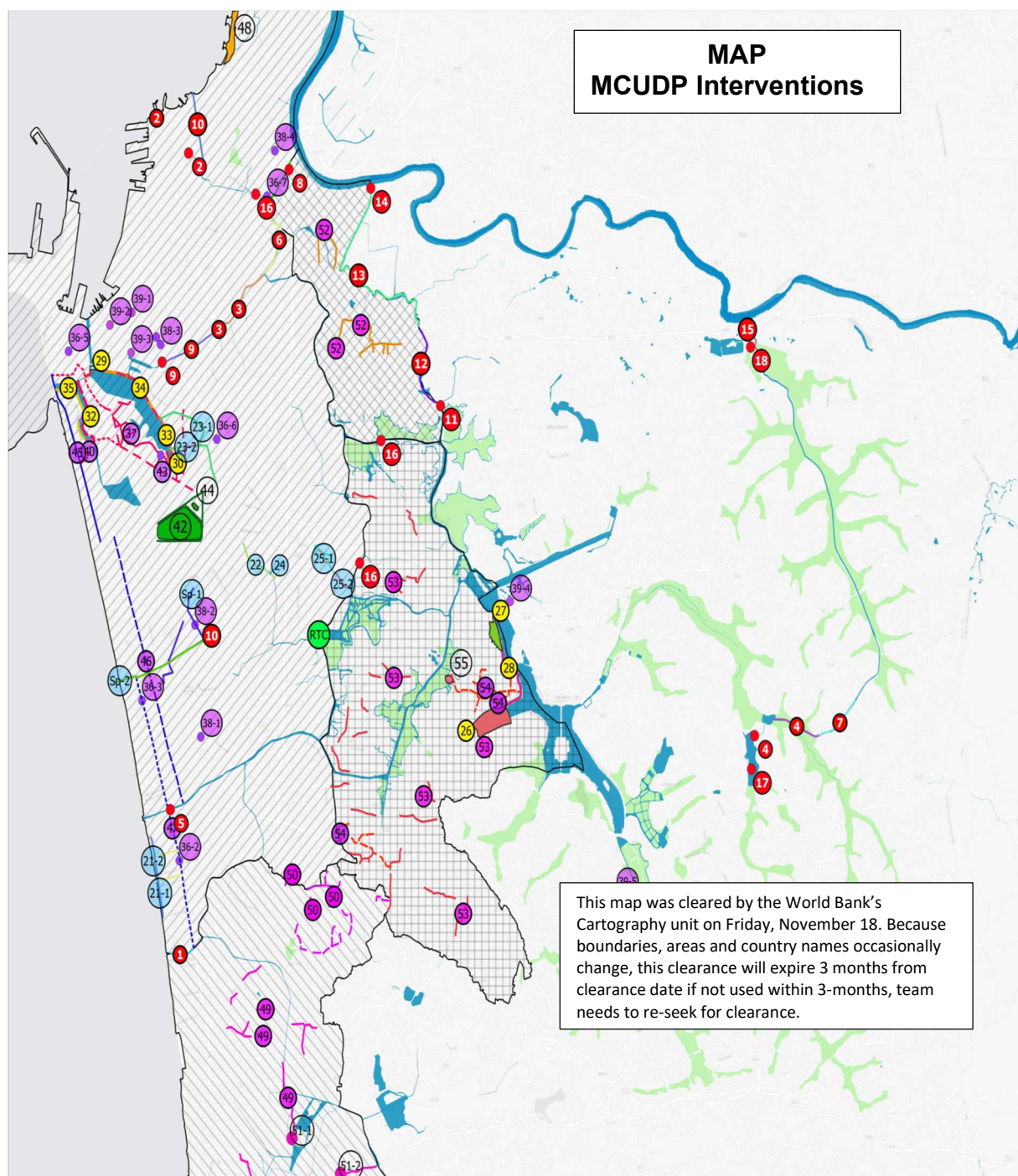


Road, Sidewalks, and Local Drainage Improvement



Nature-Based Solutions and Wetland Parks







Legend

Component 1.1

- 1 - Canal bank protection work of Dehiwala Canal
- 2 - Aluth Mawatha Culvert, Mutwal Box Drain and Ext. Mutwal outfall
- 3 - Rehabilitation of St. Sebastian South Canal
- 4 - Improvements to Madiwela East Diversion Scheme - I
- 6 - Construction of Bank Protection works of St. Sebastian North
- 7 - Improvements to Madiwela East Diversion Scheme - II
- 8 - Design & building of St. Sebastian North Lock Gates & Pumping Station
- 9 - Design & Building of St. Sebastian South Stormwater Pumping Station
- 10 - Design & Building of New Mutwal Tunnel
- 10 - Design & Building of Torrington Tunnel
- 12 - Kolonnawa Canal Diversion Scheme Stage - II
- 13 - Kolonnawa Canal Diversion Scheme Stage - III
- 15 - Improvements to Madiwela East Diversion Scheme - III
- 18 - Design & Building of Ambathale Stormwater Pumping Station
- 2 Aluth Mawatha Culvert, Mutwal Box Drain and Ext. Mutwal outfall
- 4 Improvements to Madiwela East Diversion Scheme - I
- 5 Conveyance improvements to Wellawatta Canal at Galle Rd Bridge
- 8 Design & building of St. Sebastian North Lock Gates & Pumping Station
- 9 Design & Building of St. Sebastian South Stormwater Pumping Station
- 11 Kolonnawa Canal Diversion Scheme Stage - I
- 14 Kolonnawa Canal Diversion Scheme Stage - IV
- 16 Installation of Flushing Gates - Kolonnawa
- 16 Installation of Flushing Gates - Port Access Road
- 16 Installation of Flushing Gates - Rajagiriya
- 17 Dredging of Thalagama Tank
- 18 Design & Building of Ambathale Stormwater Pumping Station

Component 1.2

- 21-1 Marine Drive Existing Drain
- 21-2 Marine Drive Sea Outfall
- 22 Rehabilitation of Gregory's Canal
- 23-1 Norris Canal Existing
- 23-2 Norris Canal Drain Completed
- 23-3 Norris Canal Drain Not completed
- 24 Improvement to Kinsey Road Drain
- 25-1 Devi Balika Existing Drainage
- 25-2 Devi Balika Dredging Only
- 25-3 Devi Balika Improved Drainage
- Sp-1 Torrington Tunnels Secondary Network
- Sp-2 Torrington Main Tunnel

Component 1.3

- RTC

Component 1.4

- 26 Beddagana Bio Diversity Park
- 27 Kotte Rampart
- 29 McCallum Entrance Park
- 28 Cycle Track Nippon Mawatha
- 30 Bank Protection of Eastern Bank of East Beira Lake
- 31 Bank Protection of Western Bank of West Beira Lake
- 32 Bank Protection of Eastern Bank of West Beira Lake
- 33 Linear Park I
- 34 Linear Park II
- 35 Linear Park III

Component 2.1

- 36 Public Conveniences Pkg-1
- 38 Public Conveniences Pkg-2
- 39 Public Conveniences Pkg-3
- 37 Walkability Improvement Pk 1
- 40 Walkability Improvement Pk 2
- 43 Walkability Improvement Pk 3
- 41 Model Zone Development in Town Hall - Package II
- 42 Model Zone Development in Town Hall - Pkg II
- 44 Model Zone Development in Town Hall - Pkg I
- 45 Galle Road+R. A. De Mel Mawatha - Pkg A
- 46 Galle Road+R. A. De Mel Mawatha - Pkg B
- 47 Galle Road+R. A. De Mel Mawatha - Pkg C
- 48 Construction of Beach Park at Crow Island
- 49 Infrastructure Development in DMMC - Pkg I
- 50 Infrastructure Development in DMMC - Pkg II
- 51-1 Vehicle Hangar DMMC - Pkg III
- 51-2 Building DMMC - Pkg I
- 52 Infrastructure Development in KUC - Pkg I
- 53 Infrastructure Development in SJKMC - Pkg I
- 54 Infrastructure Development in SJKMC - Pkg II
- 55 Angampitiya ground KJKMC Pkg III

Municipal Councils

- Colombo Municipal Council
- Dehiwala Mount Lavinia Municipal Council
- Kolonnawa Urban Council
- Sri Jayawardanapura Kotte Municipal Council
- Existing Water Bodies
- Canals and Marshes